

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

QUIZ 3 SET A

SUMMER SEMESTER, 2024-2025

DURATION: 25 MINUTES

FULL MARKS: 20

CSE 4205: Digital Logic Design

Programmable calculators are not allowed.

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

ID: _____

1. Design a $2^3 \times 3$ priority encoder with a valid output. The encoder has 8 inputs labeled I_0 to I_7 , 3 output lines labeled O_2 , O_1 , O_0 , and a valid signal V .
The inputs have the following priority order from highest to lowest:
- 5
(CO5)
(PO3)

$I_7 > I_4 > I_1 > I_0 > I_3 > I_2 > I_5 > I_6$

1. If multiple inputs are HIGH, the encoder outputs the binary code of the highest-priority input.
2. The valid output V is 1 if any input is HIGH, and 0 if all inputs are LOW.
3. You may arrange the inputs in the truth table in any clear order.

Complete the truth table based on this priority.

Inputs								Outputs			
								O_2	O_1	O_0	V

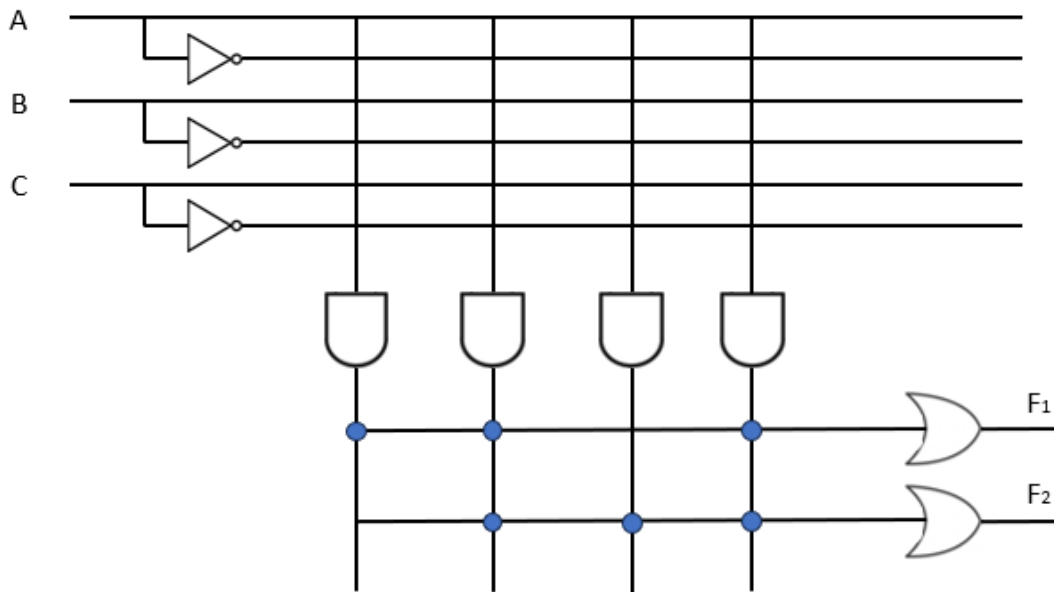
2. In the following PLD, only the AND array is programmable; the OR array is fixed. The PLD has 3 inputs, A , B , and C , and 2 outputs, F_1 and F_2 . The functions are defined as follows:

5
(CO5)
(PO3)

$$F_1(A, B, C) = \sum m(1, 2) + d(0), \quad F_2(A, B, C) = \sum m(1, 2, 4) + d(0).$$

Make appropriate connections to implement the functions F_1 and F_2 using the structure shown below. Please note that:

- No structural changes are allowed in the PLD.
- You may not change the positions of the inputs or outputs.
- Mark programmable connections with $\times$. A $\odot$ denotes a fixed connection. Intersecting wires are not connected unless one of these symbols is shown at the junction.
- Simplify the functions as needed, and then implement F_1 and F_2 in the given PLD while following the rules above.



3. In the following circuit, the inputs A , B , and C are n -bit numbers, and the output F is also an n -bit number. 6
(CO3)
(PO2)

The comparator circuit compares two n -bit numbers M and N and produces a comparison status indicating whether

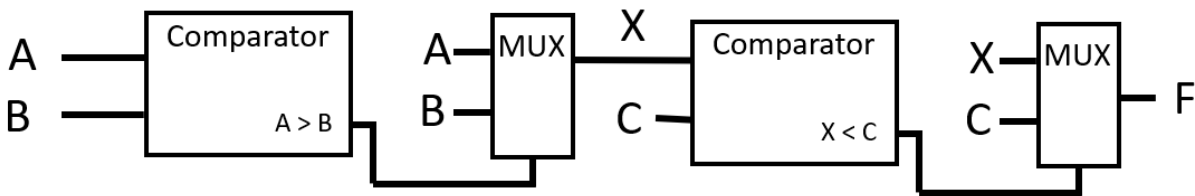
$$M > N, \quad M < N, \quad \text{or} \quad M = N.$$

Each comparator output is HIGH when its corresponding condition is satisfied. For example, $M > N$ is 1 when M is greater than N , and 0 otherwise.

The MUX is an n -bit multiplexer that selects one of its two inputs based on the selection bit. If the MUX receives M and N as inputs and the selection line is 1, it selects the first input, M .

Based on the given circuit, determine what F represents: the **largest**, **smallest**, **median (middle) value**, or something else among the three input numbers A , B , and C .

Justify your answer with a suitable example.



4. Answer the following questions:

4 × 1
(CO3)
(PO2)

- a) Suppose you are multiplying a 3-bit number by a 7-bit number. How many adders are required? (The adders may have any number of bits.)

Answer: _____

- b) How many 3×8 decoders are needed to implement a 7×128 decoder?

Answer: _____

- c) What type of PLD is shown in Question 2?

Answer: _____

- d) In a BCD adder, would an error occur if 6 were added to the preliminary sum only when the addition produced a fifth-bit carry? Explain in one sentence.

Answer: _____

[Bonus: 2] Design a circuit that takes a 4-bit binary number as input and outputs 1 if the number is odd and 0 if it is even. Mark Deduction for complicated solution.